EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**In an undirected graph, A–B is normally stored only under A in a basic adjacency list.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

### GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 053 · Graphs: Adjacency Lists and Traversal

Name: _____ Class: _____ Date: _____

**1. In a BST, where are keys smaller than the parent?**

**2. What is a graph vertex?**

**3. How does a queue differ from a stack?**

**4. An undirected graph has vertices A, B, C, D and edges A–B, A–C, C–D. In vertex order A,B,C,D, write row C of its adjacency matrix and explain why the matrix is symmetric.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

In the left subtree.

### 2. Model answer

Node representing a point/entity.

### 3. Model answer

A queue is first in, first out (FIFO); a stack is last in, first out (LIFO).

### 4. Model answer

Row C is [1,0,0,1], because C connects to A and D. For each undirected edge, both corresponding matrix cells are 1.

### By the end, you can…

• An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

• For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

• Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

### Vocabulary — know these terms

graph — Vertices linked by edges.

adjacency list — Neighbours indexed by vertex.

BFS — Breadth-first search using queue.

visited — Vertices already discovered.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why DFS can use a stack while BFS uses a queue and why cycles require visited tracking. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Write adjacency lists for undirected edges A-B,A-C,B-D. [6]

### Question 2

Trace BFS from A in the lesson graph, recording the queue after expanding A and B. [6]





NEW LEARNING

# Read, annotate and explain

### Discover

An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

### Trace

For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

### Evaluate

Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].**

**Explain the trace**

With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve